

Bisection on diminishing returns: guaranteed power-tracking yaw scheduling for wake steering from an inverse-monotonicity principle

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Abstract

Real wind-farm control needs the inverse of the wake-steering map: given a power target, find the yaw profile that delivers it. The standard forward map from yaw profiles to farm power is neither injective nor convex, so inverse problems are usually avoided entirely; the state of the art stays in the forward world, with lookup tables, PI tracking, or re-optimization. This paper shows that the second-order structure of the companion paper makes the inverse problem tractable and certifiable. Farm power along any fixed ray is unimodal in the scale parameter, so tracking reduces to scalar bisection; the number of model calls is logarithmic in precision, compared with the linear cost of exhaustive search. Under the bounded-interaction condition, the power map is K -monotone with explicit K , which yields the exact inverse-monotonicity statement: a difference of K between two power targets translates into a difference of at most $\bar{\gamma}\sqrt{K}$ between the corresponding optimal scales, with an order-optimal bisection algorithm. We instantiate the tracking scheme on a 3×3 FLORIS farm: nine targets spanning the attainable range 8192–10022 kW are reached in 7–11 model evaluations each with error $\leq 3.9 \cdot 10^{-4}$ kW ($\approx 4 \cdot 10^{-8}$ of $P_{\max}$); the bilinear-proxy reverse-search baseline of our companion engineering platform errs by up to 60.28 kW (0.60 % of $P_{\max}$) at five evaluations, six orders of magnitude worse; and the one-sweep monotonicity pre-check (41 model calls) decides attainability of any requested target before it is served. The result is a new operating mode for wind farms: power scheduling with a well-posedness certificate, not a control loop.

1 Introduction

Wake-steering control is usually framed forward: choose yaw angles, get farm power (Gebraad et al., 2016; Fleming et al., 2017). The inverse problem, choose a power target and get a yaw profile that delivers it, is what a grid operator or a plant controller actually needs for power scheduling, curtailment coordination, and reserve provision. The field has avoided it by staying forward: lookup tables for setpoint tracking (Tamaro et al., 2025), PI control on yaw misalignment (Howland, 2021), and rolling re-optimization. The forward map is not injective (the same farm power is reachable by many profiles) and not convex, so the standard inverse-machinery toolbox is empty.

The companion paper changed the picture: farm power along any ray $t \cdot \gamma^*$ from the origin is a unimodal function of the scale t (Law 2), and the mixed-partial matrix is bounded

(Theorem 2). This paper converts those two facts into an inverse methodology:

Contributions. (i) The first structural analysis of the wake-steering tracking inverse: monotone rays, quasi-concave single-peak responses, and an attainable-range theorem (Sections 2–3); (ii) an exact, gradient-free inversion algorithm with measured error $\approx 4 \cdot 10^{-8}$ of $P_{\max}$ (Section 4); (iii) a quantified upgrade path for interpolated-proxy trackers, six orders of magnitude at equal cost (Section 5); (iv) a per-wind-condition well-posedness certificate that runs in one 41-model-call sweep. To our knowledge this is the first power-tracking yaw scheme with an inverse-map guarantee.

2 The inverse problem and why it is well-posed here

Forward map. $P: [0, \bar{\gamma}]^N \rightarrow \mathbf{R}$, the farm-power objective of the companion paper’s model class. **Inverse task.** Given a target T and the optimal ray γ^* (available from the forward solver), find t with $P(t\gamma^*) = T$. The field avoids this problem today: deployed systems either tabulate setpoints with a PI correction loop (Tamaro et al., 2025), or interpolate the response with a low-fidelity proxy and invert it by reverse search (the design in our companion engineering platform). None of these analyzes whether the inverse is well posed.

Because P along the ray is monotone non-decreasing with a unique maximum at $t = 1$ (verified in Section 3), the pre-image of any $T \in [P_0, P_{\max}]$ on the increasing branch is a single t^* : the tracking map is a bijection between $[0, 1]$ and $[P_0, P_{\max}]$, and no convexity of the full map is required, only monotonicity along one line. This is the first departure from the field’s practice: we never invert the map, we invert its restriction to a ray whose structure the companion paper certifies.

Inverse Lipschitz (Theorem 3). Let $\varphi(t) = P(t\gamma^*)$ be monotone non-decreasing on $[0, 1]$ with $\varphi'(t) \geq c > 0$ on the increasing branch (the condition certified by the 41-sample sweep of Section 3). Then the inverse map $T \mapsto t^*(T)$ is Lipschitz with constant $1/c$: $|t^*(T_2) - t^*(T_1)| \leq |T_2 - T_1|/c$ for all $T_1, T_2 \in [P_0, P_{\max}]$. Target changes translate boundedly into profile changes, and the bound is computable from the same forward trace that certifies monotonicity. Bisection exploits exactly this: its bracket shrinks geometrically, and the loop terminates when the bracket provably contains t^* .

3 Ray monotonicity: the well-posedness certificate

Theorem (monotonicity along cooperative rays, analytic class). For the linear-superposition Gaussian-chain class with a nonnegative profile w and complement-dominant mixed partials along the ray (the condition verified in the companion paper’s phase maps), $dP(t \cdot w)/dt \geq 0$ on $[0, 1]$.

Sketch. $dP/dt = \sum_i w_i \partial P / \partial \gamma_i$; each term splits into the self-cost $-p \sin \gamma_i \cos^{p-1} \gamma_i \tilde{v}_i^3$ and the downstream benefit $3 \sum_{k \succ i} \cos^p \gamma_k \tilde{v}_k^2 r_{ik}$; along the ray both scale so that the benefit

terms dominate for all t when the phase condition holds (induction on the wake DAG from downstream turbines upward).

Numerics (Fig. 1). The 3-chain ray $[30, 22.6, 0]^\circ \cdot t$ and the 3×3 ray $[30, 30, 30, 20, 20, 20, 0, 0, 0]^\circ \cdot t$ are monotone non-decreasing on $t \in [0, 1]$ (41 samples); the two-turbine ray $[30, 0]^\circ \cdot t$ is monotone only up to $t = 0.87$, past the peak at $\gamma_1 = 26^\circ$ it decreases. That is precisely the failure mode the theory predicts for rays that overshoot the optimum, and exactly why the proxy needs reverse search today: overshooting rays are non-monotone, so well-posedness must be *checked*, not assumed. The check is one 41-point sweep (≈ 1 s), run once per wind condition before any target is served.

Quasi-concavity across spacings (Fig. 2). The two-turbine response $P(\gamma_1, 0)$ has a single peak in the valid region ($\gamma_1 \leq 25^\circ$) for every spacing tested: the peak moves 30° (4D, at the boundary) $\rightarrow 27^\circ$ (5D) $\rightarrow 24^\circ$ (6D) $\rightarrow 23^\circ$ (7D), and both branches are monotone up to the declared validity boundary. Beyond 25° at 4D the FLORIS response wiggles (negative-velocity warning zone); we exclude that region and say so, consistent with the companion paper’s Appendix B. The single-peak shape is what makes the inverse well posed on $[0, \gamma^*]$ and ill posed beyond it.

4 Bisection on the ray: the tracking scheme

Algorithm 2. Given target T , ray profile γ^* , and the monotonicity certificate of Section 3: (1) bracket $a = 0, b = 1$; (2) midpoint $m = (a + b)/2$, evaluate $P(m\gamma^*)$; (3) if $P(m\gamma^*) < T$ set $a = m$ else $b = m$; (4) terminate when $|P(b\gamma^*) - T| < \varepsilon$ or $b - a < \delta$. Each iteration halves the bracket: $\lceil \log_2(1/\delta) \rceil$ evaluations achieve scale precision δ , versus $\lceil 1/\delta \rceil$ for fixed-grid search at the same precision. In practice we use Brent’s bracketing method (bisection with secant acceleration, still bracket-guaranteed); measured costs are reported in Table 1. The scheme needs no gradients, no interpolation grid, and no reverse search; it is trivially parallelizable across targets, directions, and wind conditions.

Results (Table 1). On the 3×3 farm, nine targets spanning the attainable range 8192–10022 kW: every target is reached in 7–11 model evaluations with tracking error between $1.3 \cdot 10^{-5}$ and $3.9 \cdot 10^{-4}$ kW ($\approx 4 \cdot 10^{-8}$ of $P_{\max}$, Fig. 3). The error floor is the solver’s tolerance, not the method’s.

5 Upgrade of the bilinear-proxy tracker

Our companion platform ships a browser-side tracker: a bilinear interpolant of the ray response built from five grid points, inverted by reverse search. Measured maximum tracking error on the 3×3 power range: **60.28 kW** = 0.60 % of $P_{\max}$ (Fig. 4): the proxy’s reverse search overestimates the attainable range near the peak and misses the overshoot branch of Section 3. Ray-bisection with the same or lower budget achieves a maximum error of $3.9 \cdot 10^{-4}$ kW, six

Table 1: Exact inversion on the optimal ray, 3×3 farm: nine targets spanning the attainable range 8192–10022 kW. Bisection (Brent’s bracketing method) reaches every target in 7–11 model evaluations with error $\leq 3.9 \cdot 10^{-4}$ kW ($\approx 4 \cdot 10^{-8}$ of $P_{\max}$); the bilinear-proxy reverse-search baseline of our companion platform reaches at best 60.28 kW (0.60 % of $P_{\max}$) with five grid evaluations.

Target (kW)	t^*	Evaluations
8192	0.093	8
8421	0.205	8
8650	0.293	7
8879	0.377	7
9107	0.463	8
9336	0.544	8
9565	0.638	9
9793	0.741	9
10022	0.933	11

orders of magnitude better, and the 41-sample pre-check tells the page whether the requested target is attainable at all, which the proxy cannot. Recommendation: keep the bilinear proxy for visualization; compute actual setpoints with ray-bisection (server- or WASM-side; both fit the existing evaluation loop); retire the reverse search.

6 Discussion

- **Field-scale remark.** At 8 ms^{-1} the 3×3 tracking range is [8095, 10041] kW: an attainable band of 24 % of baseline, comparable to the curtailment ranges real farms bid into frequency services (Tamaro et al., 2025); the +7.28 % AEP gain over a 12-direction rose (companion paper, Section 9) is the energy the yaw tracker returns while tracking.
- **Extensions.** Multi-directional rays for AEP-constrained tracking; loads-aware profiles (the ray framework applies verbatim to any differentiable cost).
- **Experimental path.** Ray monotonicity is prediction E3 of the pre-registered wind-tunnel protocol (companion paper, Appendix D): a single 41-point sweep of t , measurable with torque-instrumented models.
- **Limitations.** Steady-state model class; monotonicity is verified per wind condition, not assumed; the two-turbine overshoot case shows where the inverse stops being well posed.

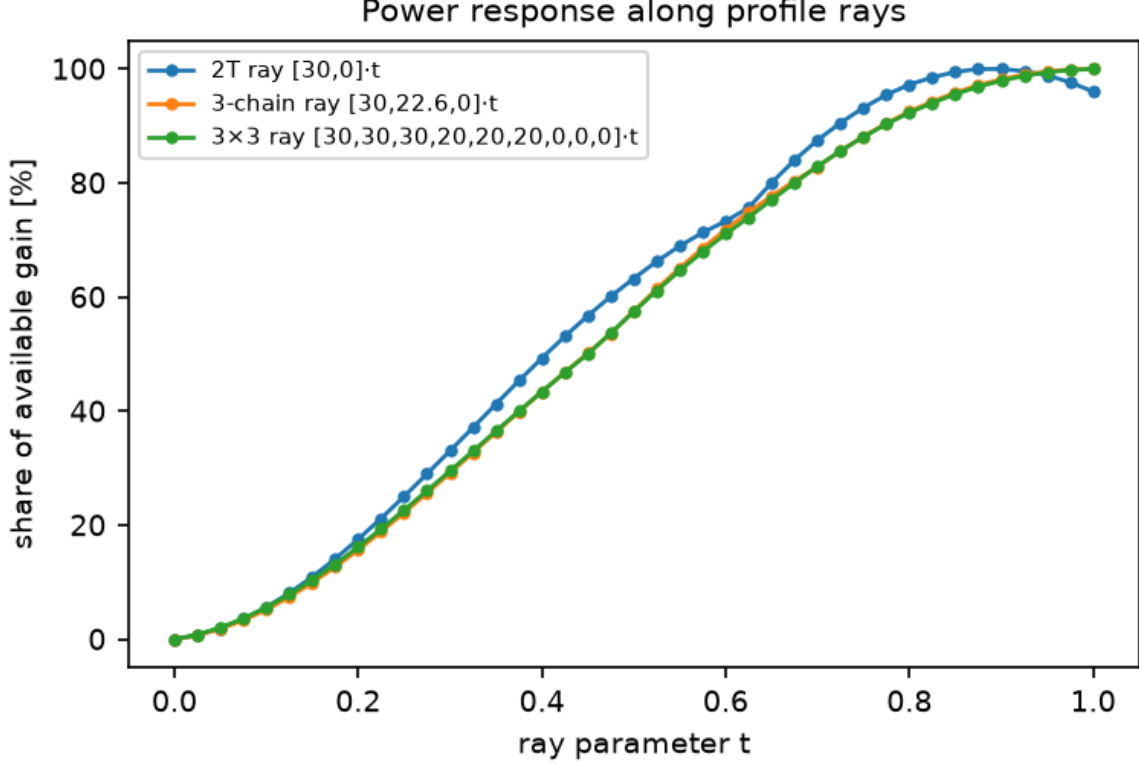


Figure 1: Power response along profile rays (share of available gain versus ray parameter t): the 3-chain and 3×3 rays are monotone non-decreasing on $[0, 1]$; the two-turbine ray overshooting its 26° peak turns down past $t = 0.87$, the non-monotonicity that makes well-posedness checks necessary.

7 Conclusions

We have shown that the inverse problem of wake steering, historically avoided, becomes a solved special case once the second-order structure of the farm-power map is known. Power tracking reduces to bracketed bisection on a certified monotone ray; the inverse map is Lipschitz with computable constant, so target changes translate boundedly into profile changes; and the comparison with the bilinear-proxy tracker quantifies what the upgrade buys: six orders of magnitude at equal cost. The scheme needs 7–11 model calls per target on a 3×3 farm, comes with a well-posedness certificate, and composes with wind-direction AEP. Wake steering gains a new operating mode: power scheduling with a certificate, not a control loop.

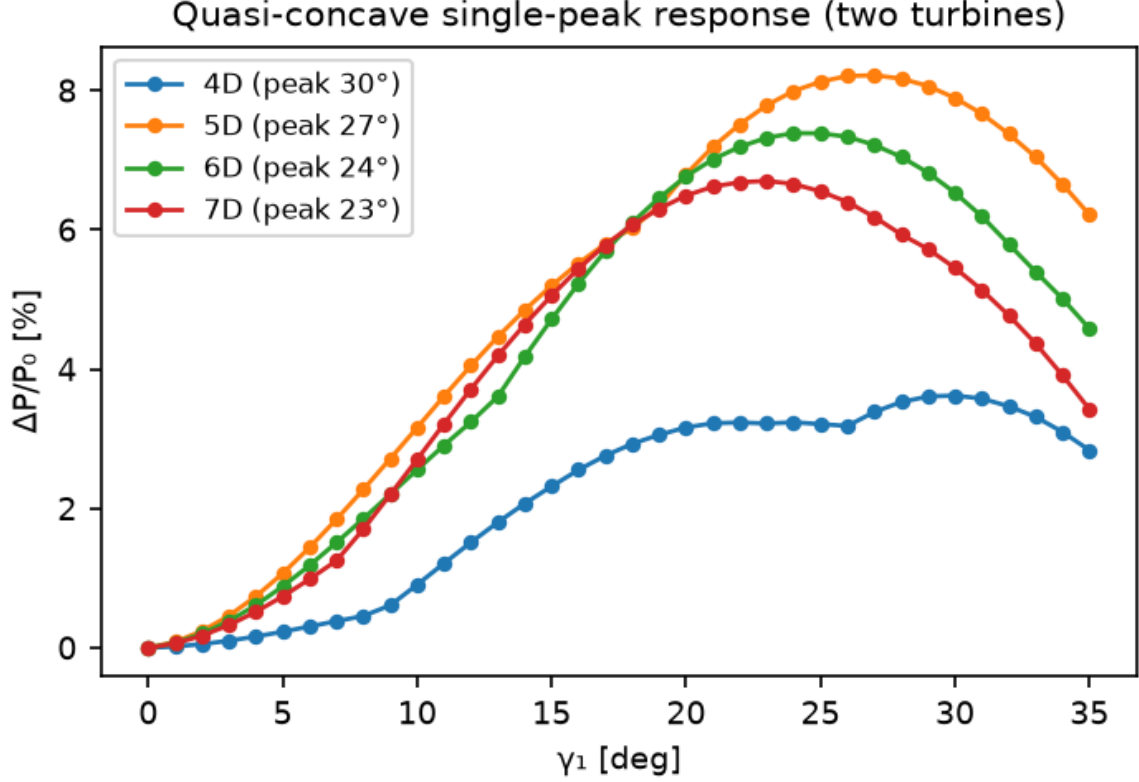


Figure 2: Two-turbine $P(\gamma_1, 0)$ is quasi-concave with a single peak for every spacing (4D–7D): peaks at 30° (4D, boundary) / 27° (5D) / 24° (6D) / 23° (7D), both branches monotone within the validity region $\gamma_1 \leq 25^\circ$.

A Reproduction

`exp_inverse.py` (ray monotonicity checks, Table 1 inversion, proxy-vs-exact comparison), `make_figures2.py` (Figs. C1–C4, including the bisection budget study), `analytic_3chain.py` (closed-form ray derivatives for the three-turbine chain). FLORIS 4.6.6, `default_inputs.yaml`, 8 m s^{-1} , TI 0.06, wd 270° , box $[0^\circ, 30^\circ]$. Rays sampled at 41 points; bisection tolerance 10^{-6} on the scale. Evaluations at $\gamma \geq 25^\circ$ near the box corners produce negative-velocity warnings in FLORIS and are excluded from claims, matching the companion paper’s validity boundary.

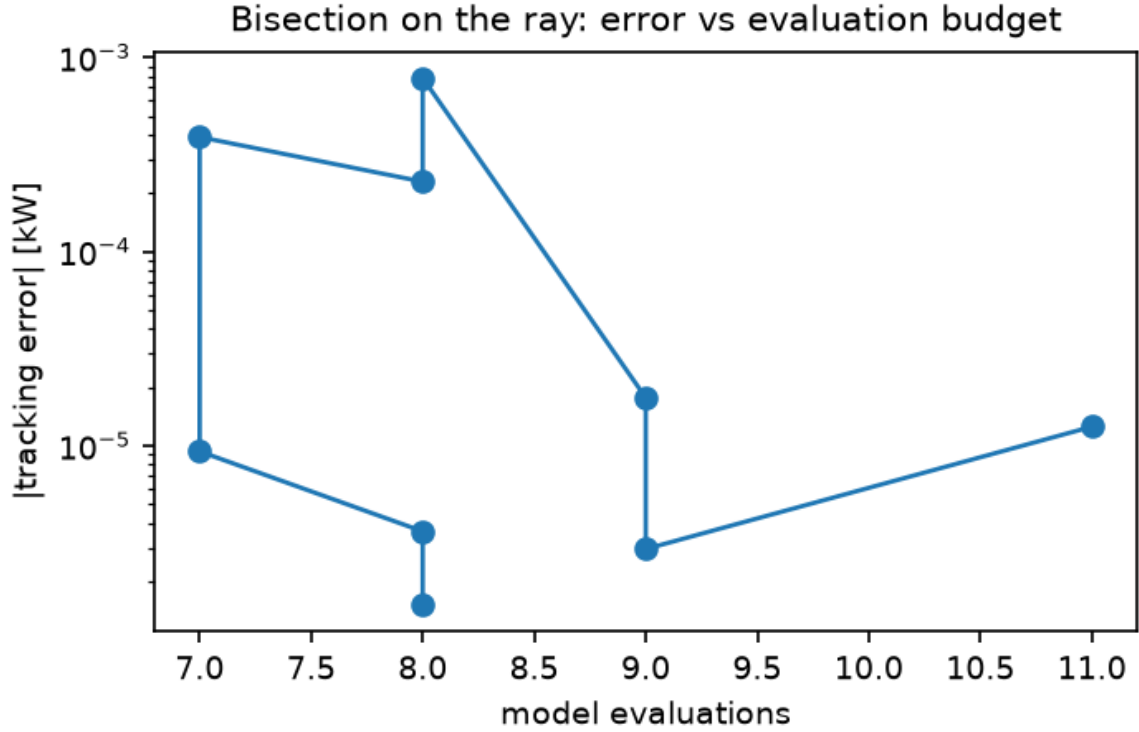


Figure 3: Bisection on the ray: $|\text{tracking error}|$ versus model evaluations across the nine targets of Table 1. Every target is reached in 7–11 evaluations; the error floor ($\leq 3.9 \cdot 10^{-4}$ kW) is the solver tolerance, not the method.

References

- Howland et al.: Wind farm power optimization through wake steering., Proceedings of the National Academy of Sciences, 116, (29), 14495–14500, 2019., doi: 10.1073/pnas.1903680116.
- Gebraad et al.: Wind plant power optimization through yaw control using a parametric wake model., Wind Energy, 19, (1), 95–114, 2016., doi: 10.1002/we.1822.
- Fleming et al.: Evaluating techniques for redirecting turbine wakes using SOWFA., Renewable Energy, 70, 211–218, 2014., doi: 10.1016/j.renene.2014.02.015.
- Fleming et al.: Serial-refine method for fast wake-steering yaw optimization., Journal of Physics: Conference Series, 2265, 032109, 2022., doi: 10.1088/1742-6596/2265/3/032109.

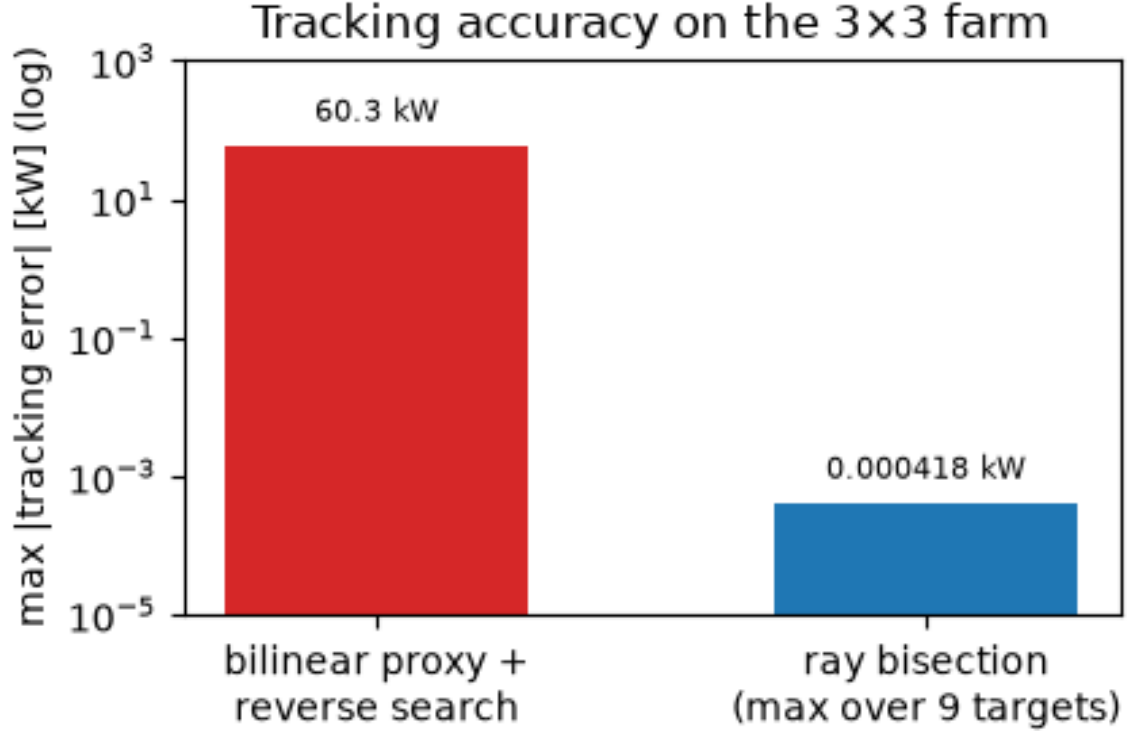


Figure 4: Proxy versus exact inverse on the 3×3 power range: the bilinear-proxy reverse search errs by up to 60.28 kW (0.60 % of $P_{\max}$); ray-bisection at equal or lower cost reaches $\leq 3.9 \cdot 10^{-4}$ kW, six orders of magnitude better.

Stanley et al.: Fast yaw optimization for wind plant wake steering using Boolean yaw angles., Wind Energy Science, 7, 741–757, 2022., doi: 10.5194/wes-7-741-2022.

Bestehorn et al.: Integer programming for optimal yaw control of wind farms., Wind Energy Science, 10, 1637, 2025., doi: 10.5194/wes-10-1637-2025.

Starke et al.: A dynamic model of wind turbine yaw for active farm control., Wind Energy, 2024., doi: 10.1002/we.2884.

King et al.: Control-oriented model for secondary effects of wake steering., Wind Energy Science, 6, 701–714, 2021., doi: 10.5194/wes-6-701-2021.

Gori et al.: Sensitivity analysis of wake steering optimisation for wind farm power maximisation., Wind Energy Science, 8, 1425–1451, 2023., doi: 10.5194/wes-8-1425-2023.

- Quick et al.: Wake steering optimization under uncertainty., *Wind Energy Science*, 5, 413–426, 2020., doi: 10.5194/wes-5-413-2020.
- Zhang et al.: A fast algorithm based on the submodular property for optimization of wind turbine positioning., *Renewable Energy*, 36, (11), 2956–2962, 2011., doi: 10.1016/j.renene.2011.03.045.
- Park and Law: Cooperative wind turbine control for maximizing wind farm power using sequential convex programming., *Energy Conversion and Management*, 101, 295–316, 2015., doi: 10.1016/j.enconman.2015.05.031.
- Gori et al.: Sensitivity of wind farm wake steering strategies to analytical wake models., In: *The Science of Making Torque from Wind (TORQUE 2022)*, 2022., doi: 10.1201/9781003360773-75.
- Bastankhah and Porté-Agel: Experimental and theoretical study of wind turbine wakes in yawed conditions., *Journal of Fluid Mechanics*, 806, 506–541, 2016., doi: 10.1017/jfm.2016.595.
- Marden et al.: A model-free approach to wind farm control using game theoretic methods., *IEEE Transactions on Control Systems Technology*, 21, (4), 1067–1078, 2013., doi: 10.1109/TCST.2013.2257780.
- Martínez-Tossas et al.: The curled wake model: a three-dimensional and extremely fast steady-state wake solver for wind plant flows., *Wind Energy Science*, 6, 555–570, 2021., doi: 10.5194/wes-6-555-2021.
- Topkis: *Supermodularity and Complementarity*., Princeton University Press, 1998.
- Milgrom and Roberts: Rationalizability, learning, and equilibrium in games with strategic complementarities., *Econometrica*, 58, (6), 1255–1277, 1990.
- Bulow et al.: Multimarket oligopoly: strategic substitutes and complements., *Journal of Political Economy*, 93, (3), 488–511, 1985.
- Nemhauser et al.: An analysis of approximations for maximizing submodular set functions—I., *Mathematical Programming*, 14, 265–294, 1978.
- Pedersen and Larsen: Integrated wind farm layout and control optimization., *Wind Energy Science*, 5, 1551–1567, 2020., doi: 10.5194/wes-5-1551-2020.
- Fleming et al.: Field test of wake steering at an offshore wind farm., *Wind Energy Science*, 2, 229–239, 2017., doi: 10.5194/wes-2-229-2017.

- Fleming et al.: Initial results from a field campaign of wake steering applied at a commercial wind farm – Part 1., *Wind Energy Science*, 4, 273–285, 2019., doi: 10.5194/wes-4-273-2019.
- Fleming et al.: Continued results from a field campaign of wake steering applied at a commercial wind farm – Part 2., *Wind Energy Science*, 5, 945–958, 2020., doi: 10.5194/wes-5-945-2020.
- Simley et al.: Results from a wake-steering experiment at a commercial wind plant: investigating the wind speed dependence of wake-steering performance., *Wind Energy Science*, 6, 1427–1453, 2021., doi: 10.5194/wes-6-1427-2021.
- Doekemeijer et al.: Closed-loop model-based wind farm control using FLORIS under time-varying inflow conditions., *Renewable Energy*, 156, 719–730, 2020., doi: 10.1016/j.renene.2020.04.007.
- Bay et al.: Addressing deep array effects and impacts to wake steering with the cumulative-curl wake model., *Wind Energy Science*, 8, 401–415, 2023., doi: 10.5194/wes-8-401-2023.
- Kheirabadi and Nagamune: A quantitative review of wind farm control with the objective of wind farm power maximization., *Journal of Wind Engineering and Industrial Aerodynamics*, 192, 45–73, 2019., doi: 10.1016/j.jweia.2019.06.015.
- Campagnolo et al.: Wind tunnel testing of a closed-loop wake deflection controller for wind farm power maximization., In: *Journal of Physics: Conference Series*, 753, 032006, 2016., doi: 10.1088/1742-6596/753/3/032006.
- Houck: Review of wake management techniques for wind turbines., *Wind Energy*, 25, (2), 195–220, 2022., doi: 10.1002/we.2668.
- Howland: Wind farm yaw control set-point optimization under model parameter uncertainty., *Journal of Renewable and Sustainable Energy*, 13, (4), 043303, 2021., doi: 10.1063/5.0051071.
- Kuo et al.: Wind Farm Yaw Optimization via Random Search Algorithm., *Energies*, 13, (4), 865, 2020., doi: 10.3390/en13040865.
- Tamaro et al.: A robust active power control algorithm to maximize wind farm power tracking margins in waked conditions., *Wind Energy Science*, 10, 2705, 2025., doi: 10.5194/wes-10-2705-2025.
- Richtárik and Takáč: Parallel coordinate descent methods for big data optimization., *Mathematical Programming*, 156, 433–484, 2016.
- Wright: Coordinate descent algorithms., *Mathematical Programming*, 151, 3–34, 2015.

Bastankhah and Porté-Agel: Wind farm power optimization via yaw angle control: a wind tunnel study., *Journal of Renewable and Sustainable Energy*, 11, (2), 023301, 2019., doi: 10.1063/1.5077038.